M4U3 · Computer vision for AECO

Architectural-items_detection

Detecting doors, windows, furniture, stairs, rooms and walls in plan views with YOLO

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github.com/cesaredellacorte-bot/Architectural-items_detection

Problem and success criteria

Problem (AECO)

- Elements on floor-plan drawings are identified and counted by hand
- Goal: a detector that recognises them automatically, as a first step towards a fast recognition workflow (quantity take-off, plan review)
- Plans are heterogeneous: scans, hand-drawn, CAD exports

Success criteria and outcome

Door, Room: mAP50 and recall ≥ 0.60 met

Window: same target not met (0.43 / 0.43)

Speed: < 0.1 s per image met (about 15 ms per image, measured in an earlier run)

Errors understood + improvement plan: done (slide 7)

Dataset

- Images: small subset (about 110) cloned from the public Roboflow Universe dataset **cad-drawings** (MIT licence)
- Labels: the six classes were **annotated by the author** in Roboflow; the original door/window/zone labels were not used
- Split: train / valid about **80 / 20** (Roboflow version 2)
- Cleaning: duplicate class *Furnitures* merged into *Furniture*; polygon rows converted to boxes; duplicate boxes removed. *Walls* kept

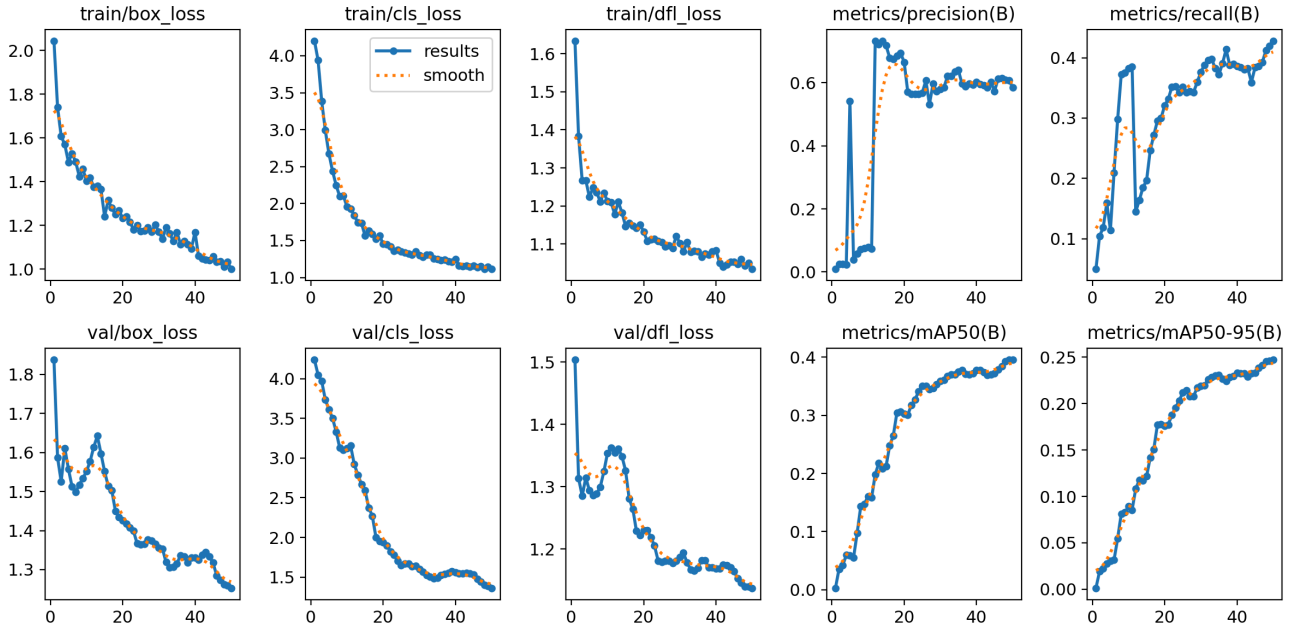
Classes (box = only the basic lines that identify the element)

ID	Class	Val. instances*
0	Door	249
1	Furniture	259
2	Room	272
3	Stairs	40
4	Walls	3
5	Window	221

*first run, before the class merge (Furniture count excludes the merged duplicate, which had 0)

Method, training and reproducibility

- **Model:** YOLOv8n, pretrained on COCO, fine-tuned
- **Training:** 50 epochs, batch 16, image size 640, seed 0
- **Software:** Ultralytics 8.4.163 (pinned)
- **Environment:** Google Colab GPU, no local install

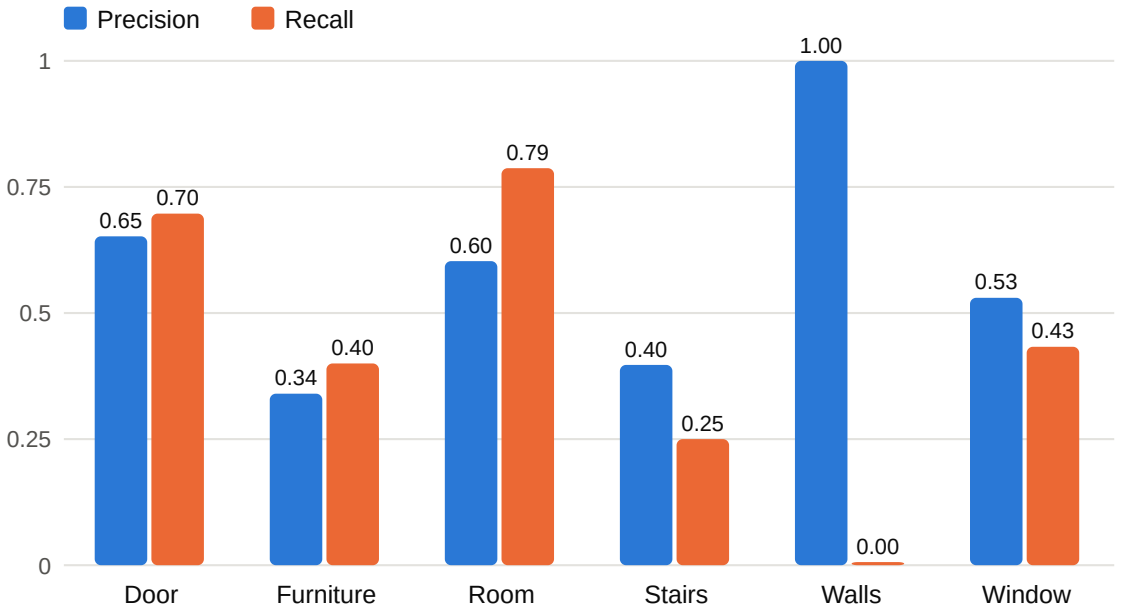


Training/validation curves of the final run: losses fall steadily; mAP50 is still rising slowly at epoch 50, precision is flat at about 0.6.

Reproduce: open **notebooks/02_train_eval.ipynb** from GitHub in Colab → Runtime: GPU → Restart session, **Run all** → metrics, curves and evidence in **results/**. Fast verification mode: 5 epochs + released weights.

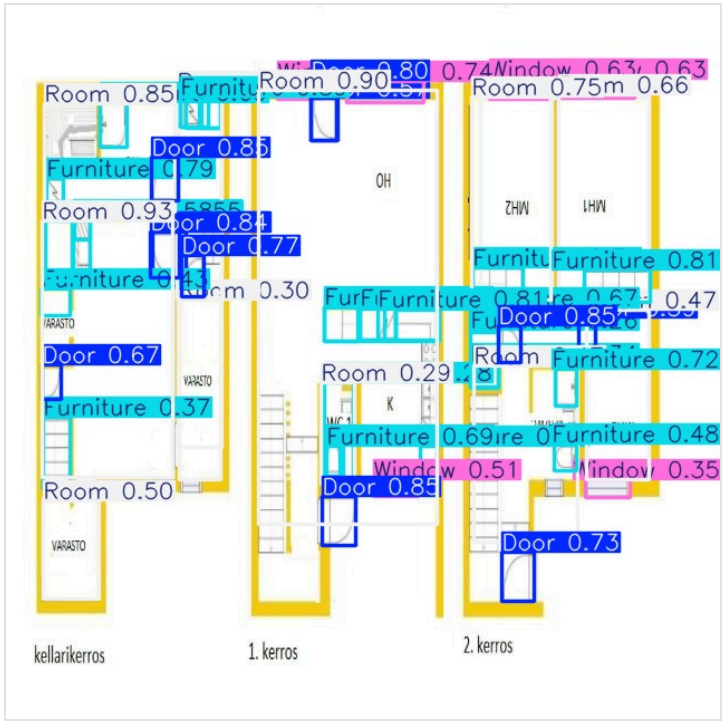
Results (validation split)

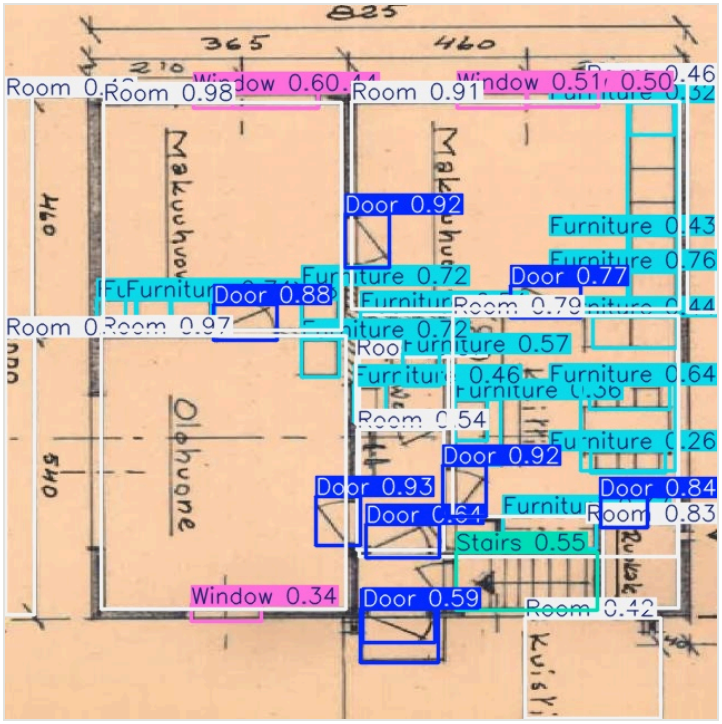
Class	Precision	Recall	mAP50	mAP50-95
ALL	**0.587**	**0.428**	**0.396**	**0.247**
Door	0.652	0.697	0.723	0.449
Furniture	0.340	0.400	0.276	0.170
Room	0.603	0.787	0.740	0.526
Stairs	0.397	0.250	0.210	0.133
Walls	1.000	0.000	0.003	0.002
Window	0.530	0.433	0.425	0.202

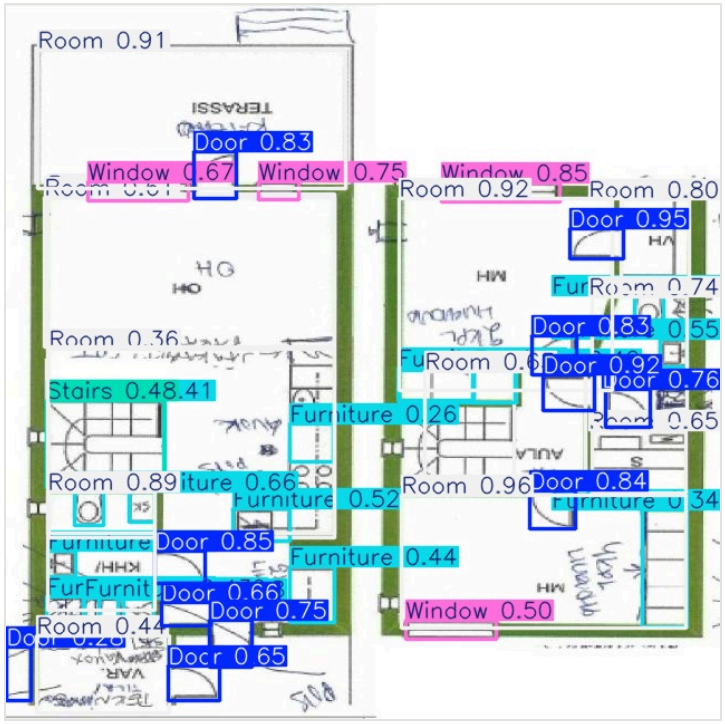


Overall: mAP50 0.40, mAP50-95 0.25. Room and Door work best (mAP50 0.74 / 0.72). Recall (0.43) is below precision (0.59): the model misses more than it invents. mAP50-95 well below mAP50 means loosely placed boxes. Walls precision of 1.0 only reflects almost no predictions.

Evidence: predictions of the final model







Reads well: doors, rooms and stairs on hand-drawn and scanned plans (Stairs 0.48-0.55). **Weak spots:** several overlapping Furniture boxes (confidence 0.3-0.8), stacked duplicate Door and Room boxes, and staircases that appear to get no box in the first plan.

Example predictions of the final YOLOv8n model on three plans of different styles. More examples (annotations, validation, new images, errors): results/evidence/ in the repo.

Error analysis and next data improvements

False positives (hypotheses)

- **Furniture** boxes stacked on one area: ambiguous symbols or unlabelled furniture
- **Window** on wall lines/dimension marks: look-alike double lines
- **Room** duplicates: large nested boxes, inconsistent room rule

False negatives (hypotheses)

- **Stairs** (recall 0.25): only 40 examples, varied styles
- **Walls** (3 examples): long thin shapes poorly fit by boxes
- **Window** in scans / hand-drawn plans (recall 0.43): low contrast, 640 px input

Next data improvements

- **1.** Audit labels, write Furniture/Room rules, add missing boxes. Target: Furniture precision 0.34 → > 0.50
- **2.** Add 40-60 stairs and hard window examples + 10-15 background images. Target: Stairs recall 0.25 → > 0.50
- **3.** Grow validation to 50+ plans, split by source plan; redefine or drop Walls

Image-level examples (red = false positive, blue = false negative): results/evidence/errors/ and docs/error_analysis.md.

Governance, licence and limitations

Governance

- Privacy: no API keys in the repo; plans are checked for personal data (owner, address) before sharing
- Minimisation: only about 110 task-relevant images
- **Risk:** a missed door or stair (false negative) is worse than a false alarm. Prefer higher recall and keep a human reviewer

Licence and data rights

- Code and annotations: MIT
- Images: cad-drawings, Roboflow Universe, MIT, cited in the README
- Weights: derived from Ultralytics (AGPL-3.0)

When not to use

Not for quantities, permits or safety decisions without human checking. Validated only on about 110 plans of limited style.

Repository: github.com/cesaredellacorte-bot/Architectural-items_detection